Design and Implementation of a Wazuh-Based SOC Monitoring, Detection and Response Platform

By

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Executive Summary

This project delivered a fully operational Wazuh-based Security Operations Center (SOC) platform deployed on a VMware virtualized environment.
The platform enables centralized visibility, real-time threat detection, vulnerability monitoring, and endpoint security telemetry for Cyberknight Security Solutions.

The implementation successfully achieved:

- Centralized SOC infrastructure
- Endpoint onboarding and telemetry ingestion
- Real-time alerting and detection
- Vulnerability and configuration monitoring
- SOC operational workflow enablement

Project Objectives

The objectives of this implementation were to establish a scalable and automated SOC platform capable of improving:

- Threat detection accuracy
- Endpoint visibility
- Incident investigation speed
- Automated threat response capability

Architecture Overview

The deployed SOC architecture consists of:

- A centralized Wazuh all-in-one server deployed as an OVA appliance on VMware.
- A Windows 11 endpoint onboarded as a monitored agent.
- A web-based Wazuh dashboard used for security operations, detection, and investigation.

The Wazuh server provides:

- Manager services
- Indexing services
- Web dashboard
- API and alert processing services

Wazuh OVA Deployment on VMware

The Wazuh platform was deployed using the official Wazuh OVA package into the VMware virtualization environment. The virtual appliance was provisioned and booted successfully and the following services were confirmed operational:

- Wazuh manager
- File Integrity Monitoring engine
- Vulnerability detection module
- Dashboard service

System access to the appliance was verified using the default Wazuh server console account.

Network Configuration and IP Address Verification

The IP address of the Wazuh server was confirmed using the Linux `ifconfig` command. This confirms that the SOC server is reachable on the internal network **Wazuh Server IP: 192.168.17.128**

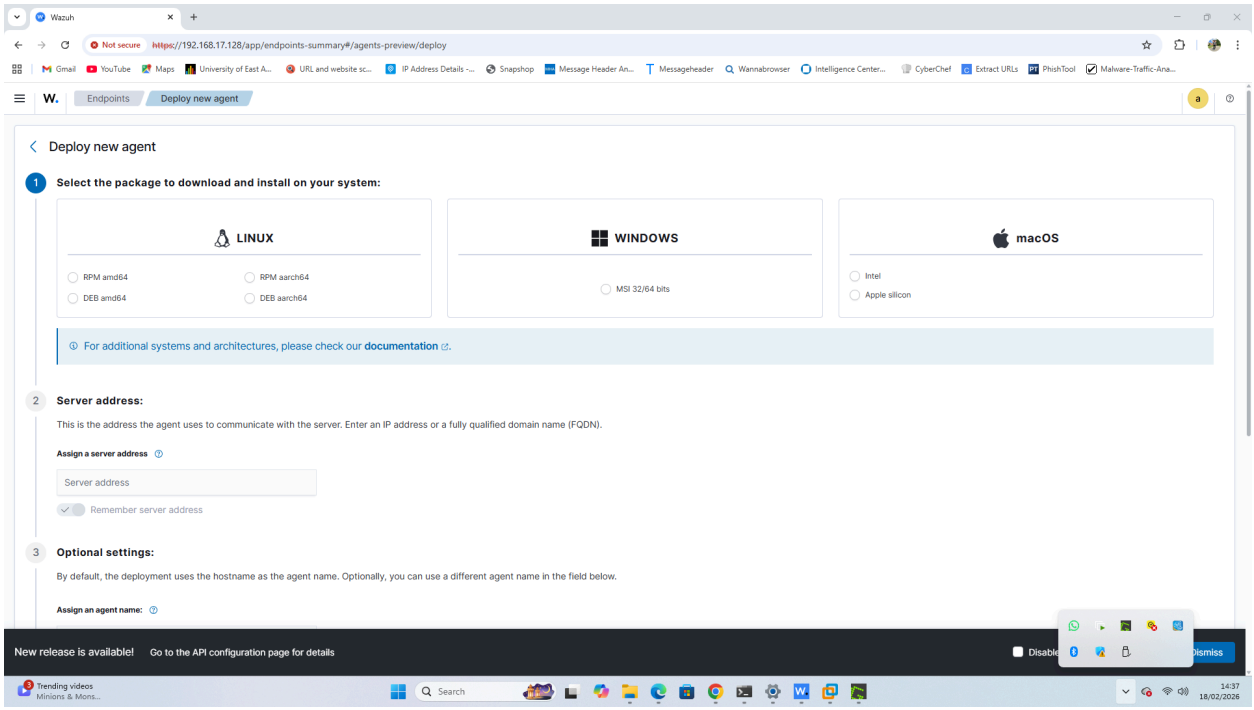
Wazuh Dashboard Access

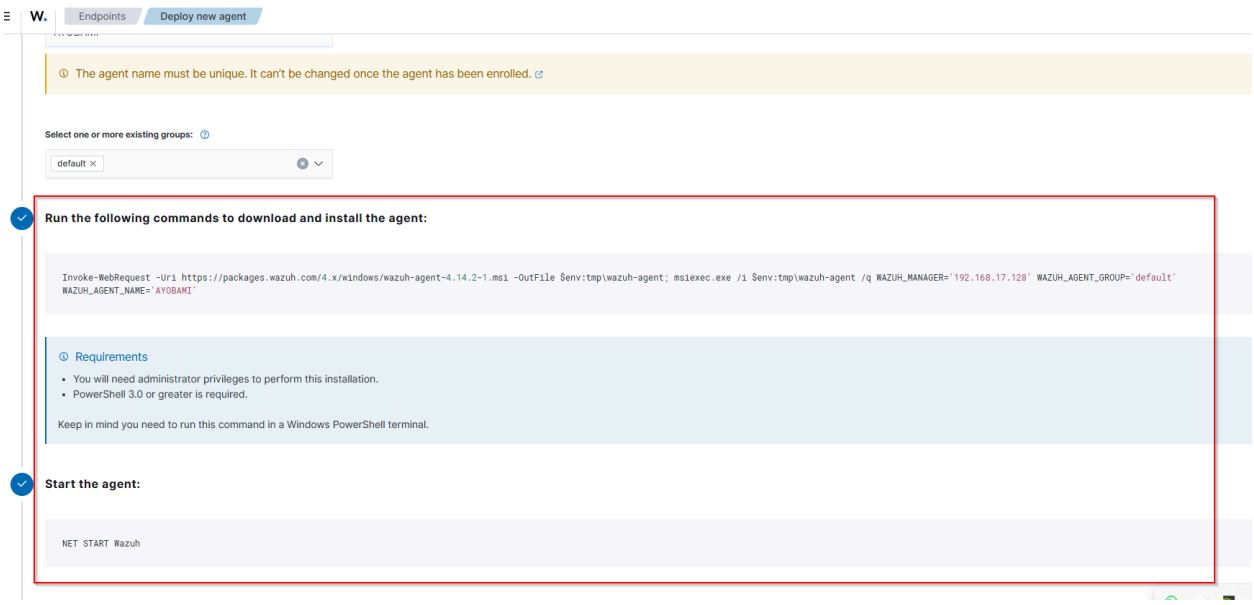
The Wazuh web dashboard was accessed using the server IP address. **Dashboard URL : https://192.168.17.128 .** The dashboard successfully loaded and authenticated, confirming:

- Web services availability
- API connectivity
- Indexing service health

Endpoint Onboarding – Host Machine Deployment

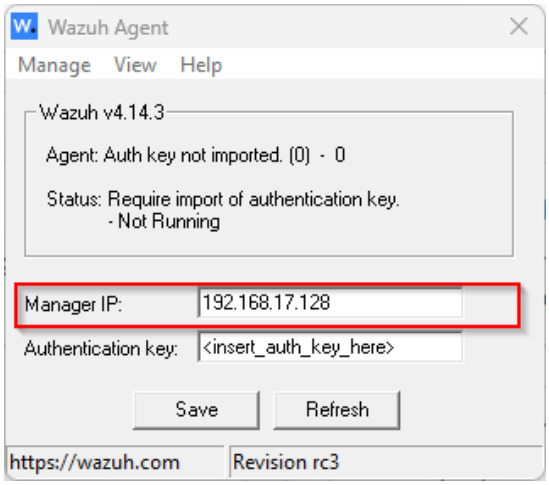
A Windows 11 Enterprise endpoint was onboarded into the SOC environment. The endpoint is successfully registered and communicating with the Wazuh manager.

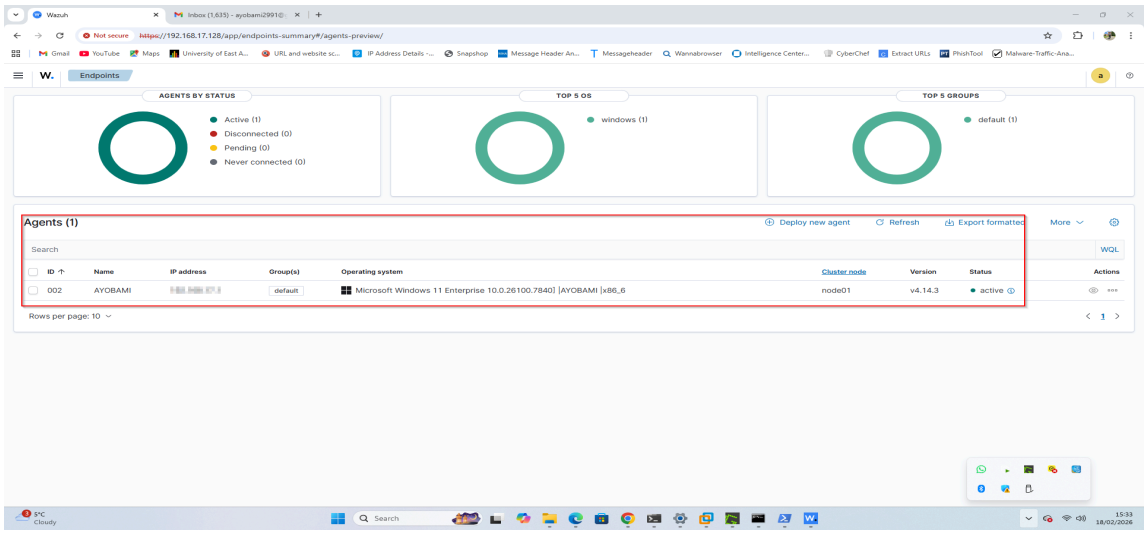




Agent Deployment Procedure (Windows Host machine)

The Windows host was onboarded using the official Wazuh agent installation workflow. The Windows agent package was downloaded and installed, followed by agent registration against the Wazuh server.





After authentication, the agent service was started and successfully established a secure connection with the manager. The dashboard confirms that the agent transitioned to **Active** status.

Security Capabilities Enabled

Real-Time Threat Detection: The endpoint is actively monitored for:

- Process activity
- Security events
- Suspicious behavior patterns

- Policy violations

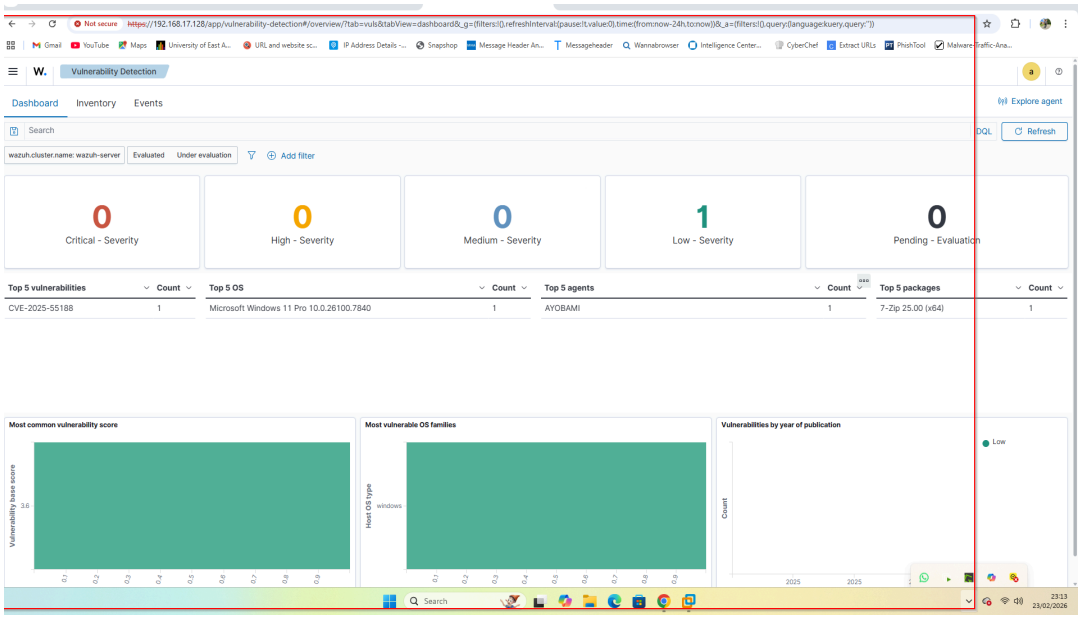
Alerts are normalized and correlated through the Wazuh rules engine.

Vulnerability Detection: The vulnerability detection module is active and continuously evaluates:

- Installed software inventory on the endpoint
- Known CVEs
- Operating system patch exposure

This enables the SOC team to identify:

- Missing patches
- High-risk software components
- Vulnerable configurations



Vulnerability details

host.os.type	windows
host.os.version	10.0.26100.7840
package.architecture	x86_64
package.name	7-Zip 25.00 (x64)
package.path	C:\Program Files\7-Zip\
package.size	0
package.type	win
package.version	25.00
vulnerability.category	Packages
vulnerability.classification	CVSS
vulnerability.description	7-Zip before 25.01 does not always properly handle symbolic links during extraction.
vulnerability.detected_at	Feb 23, 2026 @ 23:09:34.468
vulnerability.enumeration	CVE
vulnerability.id	CVE-2025-55188
vulnerability.published_at	Aug 8, 2025 @ 22:15:25.000
vulnerability.reference	https://github.com/lunbun/CVE-2025-55188/, https://youtu.be/sWT6M1cfnwM, https://lunbun.dev/blog/cve-2025-55188/, https://www.openwall.com/lists/oss-security/2025/08/09/1, https://github.com/ip7z/7zip/compare/25.00...25.01, https://sourceforge.net/p/sevenzip/discussion/45797/thread/da14cd780b/, https://github.com/ip7z/7zip/releases/tag/25.01, http://www.openwall.com/lists/oss-security/2025/08/09/1, http://www.openwall.com/lists/oss-security/2025/08/10/1, http://www.openwall.com/lists/oss-security/2025/08/13/1, http://www.openwall.com/lists/oss-security/2025/10/12/2, http://www.openwall.com/lists/oss-security/2025/10/16/7, https://www.vicarius.io/vsociety/posts/cve-2025-55188-detect-7-zip-vulnerable-version, https://www.vicarius.io/vsociety/posts/cve-2025-55188-mitigate-7-zip-vulnerability
vulnerability.scanner.condition	Package less than 25.01

The vulnerability detected on the Windows 11 endpoint is:

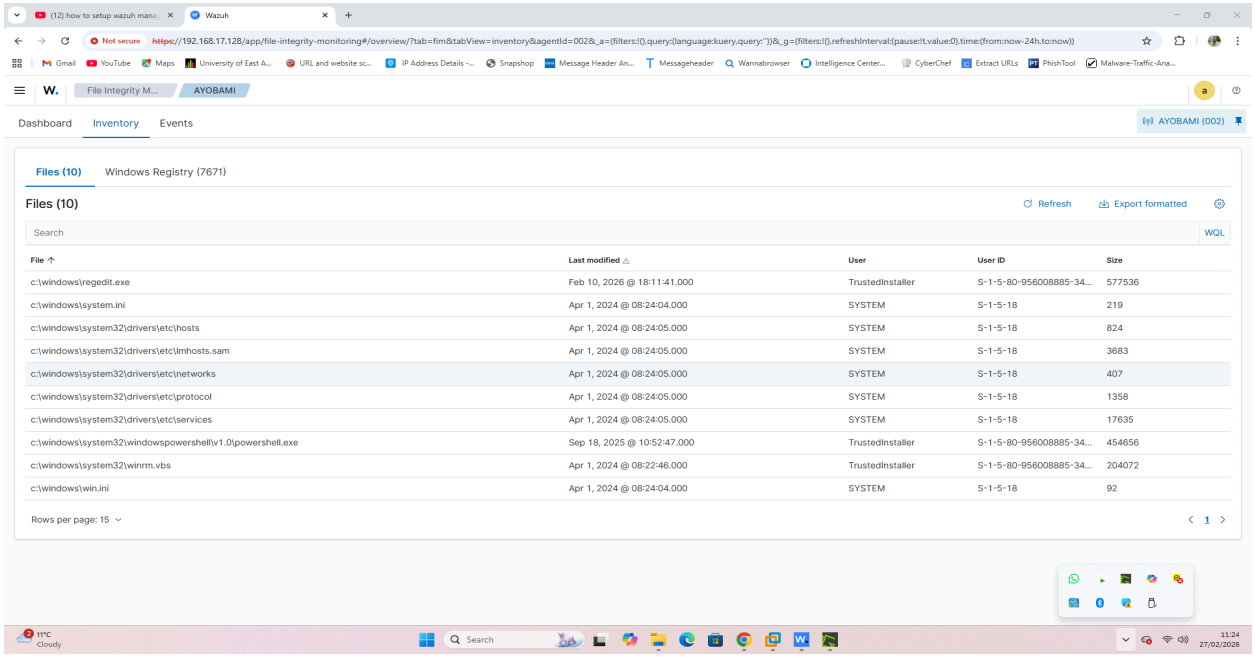
- **CVE: CVE-2025-55188**
- **Affected software:** 7-Zip 25.00 (x64)
- **Endpoint OS:** Windows 11 (10.0.26100.7840)
- **Severity (Wazuh):** Low
- Detected by Wazuh vulnerability module

The issue relates to how 7-Zip extracts files and handles symbolic links, which can allow files to be written to unintended locations when opening a specially crafted archive.

File Integrity Monitoring: The platform monitors:

- Critical system directories
- Configuration files
- Security-relevant artifacts

Any unauthorized changes generate alerts for SOC investigation.

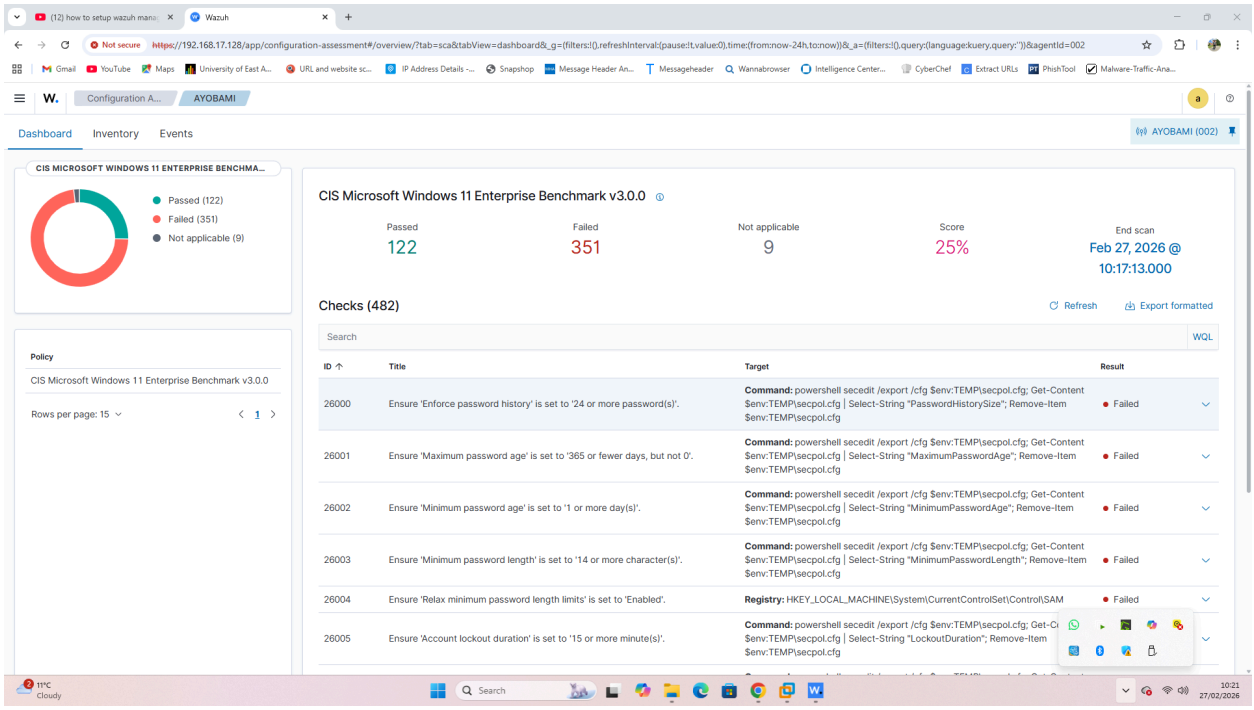


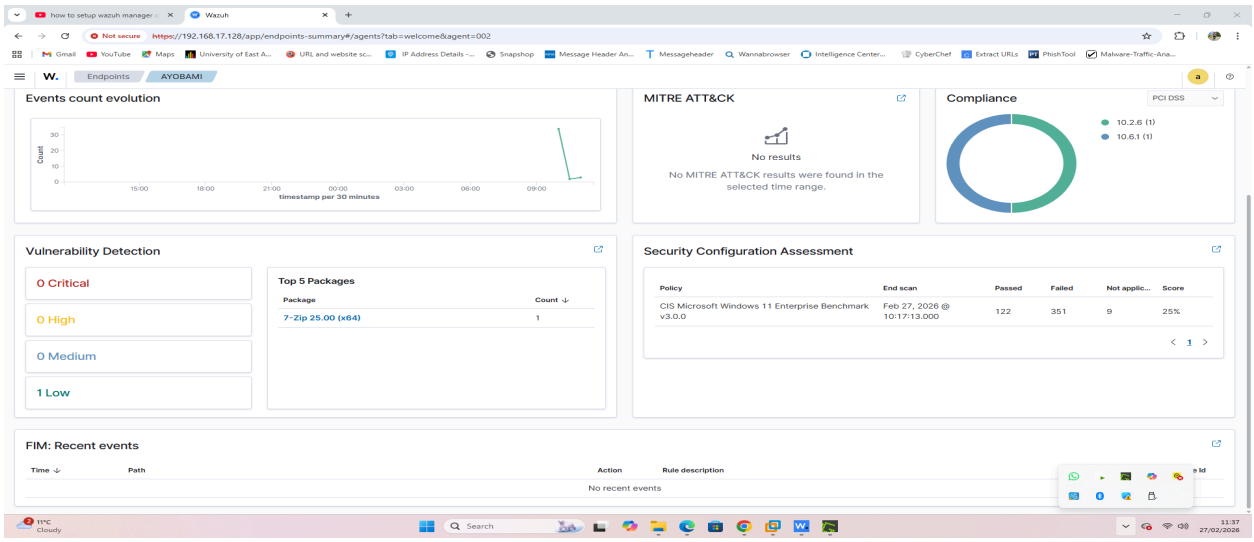
Configuration and Compliance Monitoring: The deployed configuration assessment and compliance modules support:

- Security baseline checks
- Hardening validation
- Misconfiguration detection

This supports compliance use cases such as:

- IT hygiene monitoring
- Security posture management





SOC Operational Workflow

The SOC workflow implemented in this deployment follows:

1. Endpoint telemetry ingestion
2. Real-time alert generation
3. Severity-based triage
4. Alert contextualization using endpoint metadata

5. Investigation through:
    - event timelines
    - file integrity data
    - vulnerability exposure

6. Response and containment decision-making

Security Validation Summary

The following validation checks were completed successfully:

- Wazuh server reachable on network
- Dashboard accessible and operational
- Agent authenticated and active
- Alerts being generated and processed
- Endpoint telemetry visible in SOC console
- Vulnerability and configuration modules enabled

Conclusion

The Wazuh SOC platform has been successfully deployed on VMware and operationalized to support a SOC service delivery model.

The implementation demonstrates:

- Enterprise-ready detection and response capability
- Centralized monitoring of endpoints
- Real-time security visibility
- Vulnerability and compliance oversight
- Scalable SOC architecture suitable for future expansion

This deployment provides a solid foundation for future enhancements such as:

- SOAR integration
- Threat hunting workflows
- External case management systems
- Multi-tenant SOC expansion